Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	30-09-2026
Team ID	23KD1A4233
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Commuter Route & Booking Interface	Choose Origin Station Choose Arrival Station Specify Travel Date & Time Input Passenger Count & Ticket Type
FR-2	Dynamic Fare Evaluation	Compute Distance-Based Tariff Apply Passenger Discounts / Special Rates Render Real-Time Cost Summary
FR-3	Order Processing & RITM Creation	Process Pass Order Request Assign Unique Request ID (REQ) Auto-Generate Requested Item Record (RITM)
FR-4	Pass Issuance & Verification	Render Electronic Pass Summary Display Journey & Fare Breakdown Issue Unique Ticket Identification Code
FR-5	Automated Alerts & Dispatch	Dispatch Confirmation Email Alert Trigger SMS Pass Link to Mobile Log Transmission Status in ServiceNow
FR-6	Encrypted QR Ticket Generation	Generate Unique Encrypted QR Code Create QR Code for Ticket Attach QR Pass Document to RITM Record

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability & Accessibility**	The platform shall feature a streamlined, responsive interface, enabling commuters to complete ticket purchases in minimal steps without prior system training.
NFR-2	**Security & Access Control**	End-to-end data encryption via HTTPS protocols must be enforced, alongside strict Role-Based Access Controls (RBAC) to restrict unauthorized system privileges.
NFR-3	**System Reliability & Precision**	The application must maintain high transactional integrity, delivering consistent fare computations and ticket generations with zero data corruption.
NFR-4	**Performance & Latency**	End-to-end processing—including route lookup, fare calculation, and QR pass creation—must complete within sub-3-second response times.
NFR-5	**Service Availability**	Hosted on ServiceNow cloud infrastructure, the platform must achieve 99.9% uptime, ensuring uninterrupted 24/7 access for commuters and administrators.
NFR-6	**Elastic Scalability**	The system architecture must seamlessly handle peak traffic surges during rush hours and support route network expansions without requiring infrastructure overhauls.